

# SEBIN JUNG

Mechanical Engineering Master's Student

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## INTERESTS

Safe Control, Optimization, Neural Network Verification, Reinforcement Learning, Robotics

## EDUCATION

|                                                                                                                                                                      |                                       |
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| <b>Carnegie Mellon University</b><br><i>Master of Science in Mechanical Engineering - Research</i> <ul style="list-style-type: none"><li>Total GPA 4.0/4.0</li></ul> | May. 2026 Expected<br>Pittsburgh, PA  |
| <b>Kyungpook National University</b><br><i>Bachelor of Science in Mechanical Engineering</i>                                                                         | Feb. 2024<br>Daegu, South Korea       |
| <b>University of California, Berkeley</b><br><i>Summer Session</i>                                                                                                   | Jul. 2022 - Aug. 2022<br>Berkeley, CA |

## RESEARCH EXPERIENCES

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                             |
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| <b>Intelligent Control Laboratory, CMU Robotics Institute</b> <ul style="list-style-type: none"><li>Developing safe control toolbox for humanoid robots</li><li>Integrating MuJoCo with realtime control of Unitree G1</li></ul>                                                                                                                                                                                                                                                                                            | Oct. 2024 - Present<br>Pittsburgh, PA       |
| <b>Vehicle Optimization Intelligent Control Energy Laboratory, KNU</b><br><i>Undergraduate Researcher</i> <ul style="list-style-type: none"><li>Led project on drifting control of autonomous vehicle through reinforcement learning, used CARLA for testing</li><li>Led project on implementation of autonomous driving through reinforcement learning, won <i>1st-place award</i> in graduation project competition</li><li>Participated in International University Students Autonomous EV Driving Competition</li></ul> | Jan. 2023 - May. 2024<br>Daegu, South Korea |

## HONORS AND SCHOLARSHIPS

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| <b>Outstanding Learner</b> , Hyundai Motor Group,<br>Excellent achievement award for <i>H-Mobility Class</i>                                                               | 2024 Feb.               |
| <b>1st-Place Award in Graduation Project</b> , Kyungpook National University,<br>Best research award for <i>Capstone Design of Mechanical Engineering Project 1</i> course | 2023 Spring             |
| <b>Scholarship Recommended by Dean</b> , Kyungpook National University                                                                                                     | 2022, 2023 Spring       |
| <b>Scholarship of Academic Excellence</b> , Kyungpook National University                                                                                                  | 2021 Fall - 2023 Spring |

## SKILLS

|                 |                                                                     |
|-----------------|---------------------------------------------------------------------|
| <b>Language</b> | Python, C, MATLAB, Bash, Git, $\LaTeX$                              |
| <b>Software</b> | Simulink, ROS1/2, Docker, Gazebo, MuJoCo, CARLA, IPG CarMaker, Creo |
| <b>Library</b>  | PyTorch, scikit-learn, NumPy, SciPy, matplotlib                     |

## PROJECTS

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| <b>CARLA Tutorial for Korean Students, KNU</b> <ul style="list-style-type: none"><li>Compiled and documented CARLA simulator guide for lab members and students with similar research interests</li><li>Organized into four sections: installation and execution, scenario implementation, sensor utilization, basics</li></ul>                                                                                                                                  |                                             |
| <b>Self-Driving Under Traffic Condition, KNU</b> <ul style="list-style-type: none"><li>Participated in 2024 ACC Quanser <i>Self-Driving Car Student Competition</i>, won <i>4th-place</i></li><li>Designed decision making and obstacle evasion algorithm using LiDAR scan</li><li>Utilized Pure Pursuit combined with adaptive speed control to minimize laptime</li><li>Utilized YOLOv7 for perception logic</li></ul>                                         |                                             |
| <b>H-Mobility Class, Hyundai Motor Group</b><br><i>Undergraduate Participant</i> <ul style="list-style-type: none"><li>Completed autonomous driving courses (Subjects: Perception, Decision-Making, Control, Communication)</li><li>Awarded as <i>Outstanding Learner</i></li></ul>                                                                                                                                                                              | Jul. 2023 - Feb. 2024<br>Daegu, South Korea |
| <b>Reinforcement Learning-Based Vehicle Control, KNU</b> <ul style="list-style-type: none"><li>Implemented end-to-end control of autonomous vehicle using Twin Delayed DDPG(TD3) algorithm</li><li>Designed reward functions incorporating cross-track error and heading angle error, inspired by Stanley Method</li><li>Validated algorithm under IPG CarMaker simulation</li><li>Received the <i>1st-Place Award</i> for graduation research project</li></ul> |                                             |
| <b>Autonomous Racing, KNU</b> <ul style="list-style-type: none"><li>Participated in 2023 ICCAS <i>F1Tenth Korea Championship</i></li><li>Performed SLAM based on ICP algorithm, generated optimal global path using quadratic program solver</li><li>Utilized Pure Pursuit with fine tuned speed profile for path following</li></ul>                                                                                                                            |                                             |
| <b>Electric Vehicle Fuel Efficiency Modeling, KNU</b> <ul style="list-style-type: none"><li>Developed Simulink model for accurate energy efficiency computation</li><li>Analyzed vehicle dynamics and actual power consumption data to closely match real-world driving</li><li>Reduced computation error between simulation and actual driving from 95 percent to 5 percent</li></ul>                                                                           |                                             |

## LANGUAGES

Fluent in English, Native in Korean